The Solvay Conference has gone down in history as the beginning of a famous series of debates between Albert Einstein and Niels Bohr over how to think about quantum mechanics. Bohr, a Danish physicist based in Copenhagen who is rightfully regarded as the godfather of quantum theory, advocated an approach similar to the textbook recipe we discussed in the last chapter: use quantum mechanics to calculate the probabilities for measurement outcomes, but don't ask of it anything more than that. Do not, in particular, worry too much about what is really happening behind the scenes. Supported by his younger colleagues Werner Heisenberg and Wolfgang Pauli, Bohr insisted that quantum mechanics was a perfectly fine theory as it was.

Einstein would have none of it. He was firmly convinced that the duty of physics was precisely to ask what was going on behind the scenes, and that the state of quantum mechanics in 1927 fell far short of providing a satisfactory account of nature. With his own sympathizers, such as Erwin Schrödinger and Louis de Broglie, Einstein advocated looking more deeply, and attempting to extend and generalize quantum mechanics into a satisfactory physical theory.